

PREDICTING SMALL BUSINESS SURVIVAL

Group 6:

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Introduction

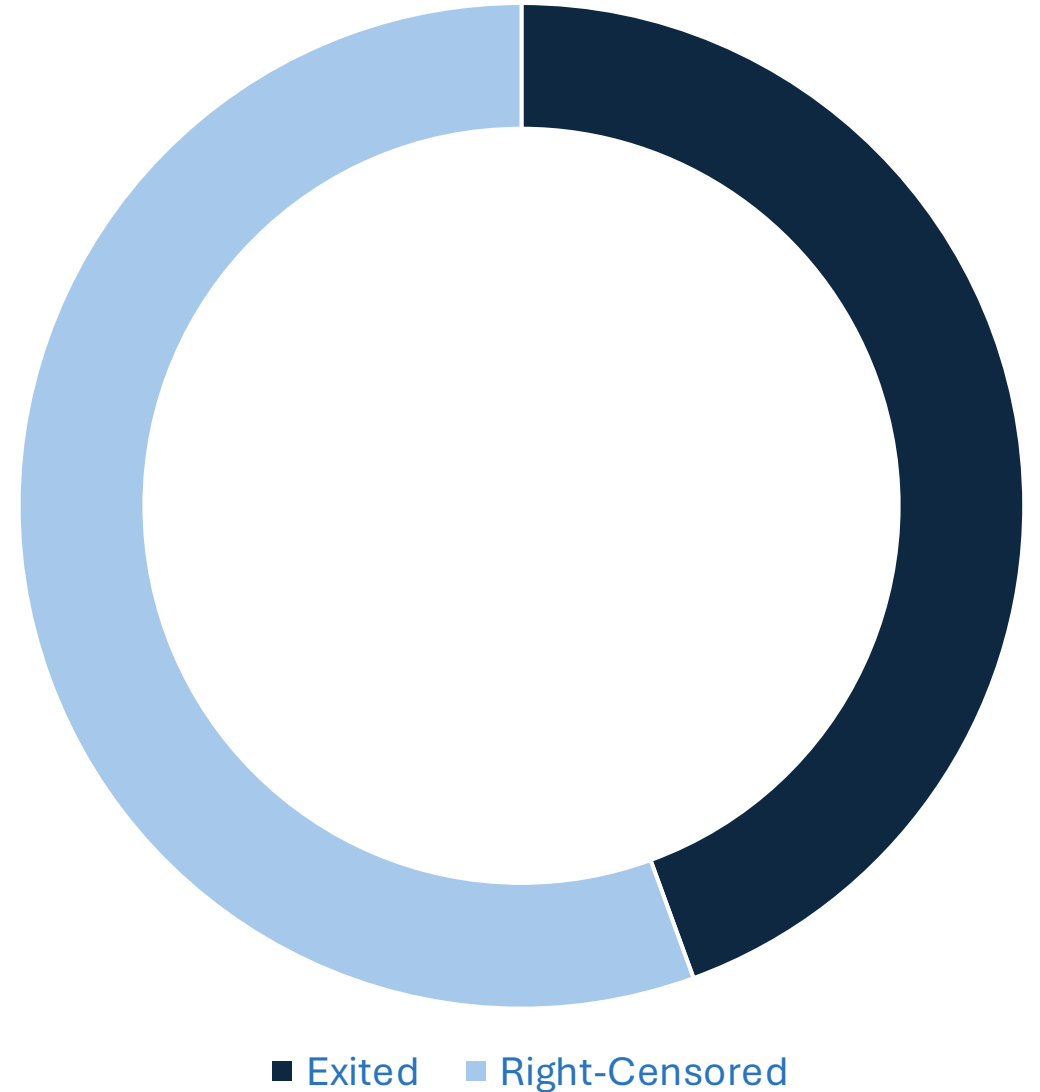
- Small businesses are essential to the U.S. economy
- **Problem:**
 - Many small businesses fail within the first few years
 - Business owners have limited tools to estimate long-term survival
 - Influenced by hundreds of changing financial and operational factors

Our Model

- **Gives us:**
 - Predicted survival rate of a small business
 - Variables with the most association with survival
- **LASSO**
 - Select important predictors
- **Cox Proportional Hazards Model**
 - Evaluate association of selected predictors with survival time
 - Predictive survival curve for a small business based on its predictor values

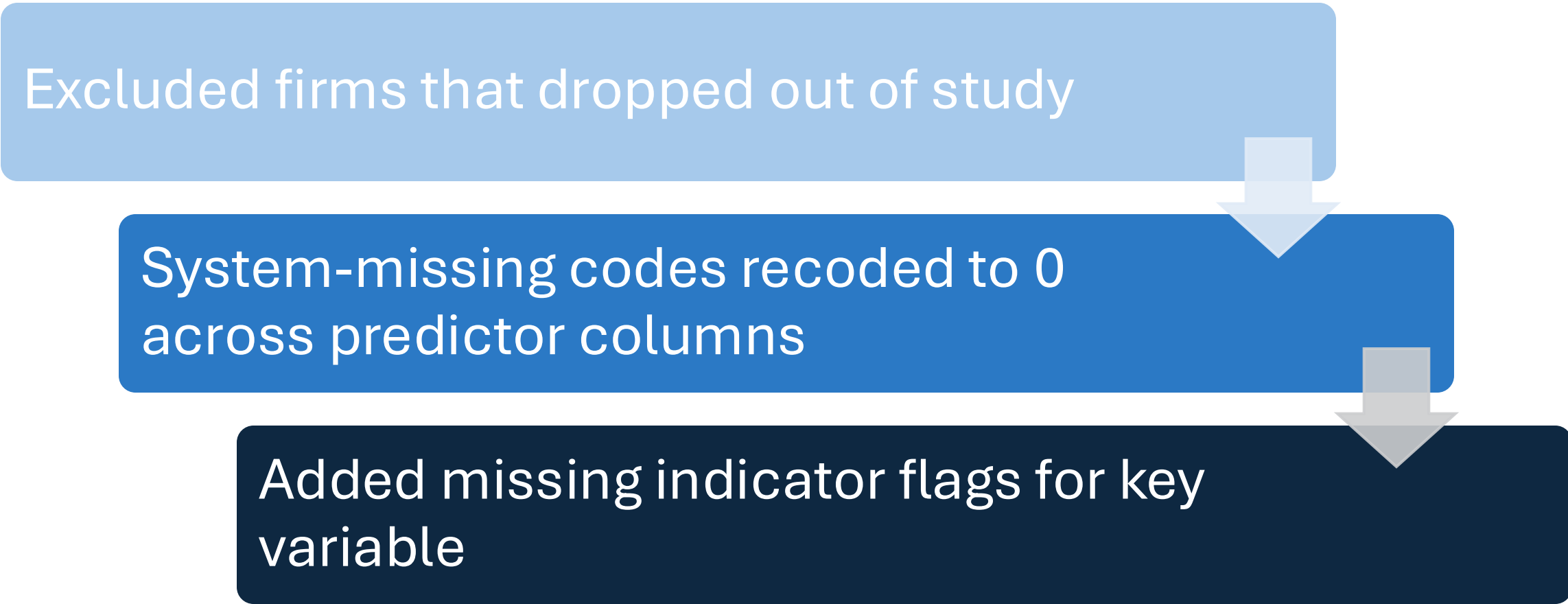
Raw Dataset

- **Kauffman Firm Survey**
 - Studies done across sectors
 - Random sampling from all new businesses included in the D&B database and reported by D&B as having started in 2004
 - 4928 firms; 208 variables
- **Time Period:** (Waves) 0-7; 2004-2011
- **Target Variable:** Duration, Event
- 2190 Exited; 2738 Right-Censored (Survived the Study)



Data Cleaning Process

Excluded firms that dropped out of study



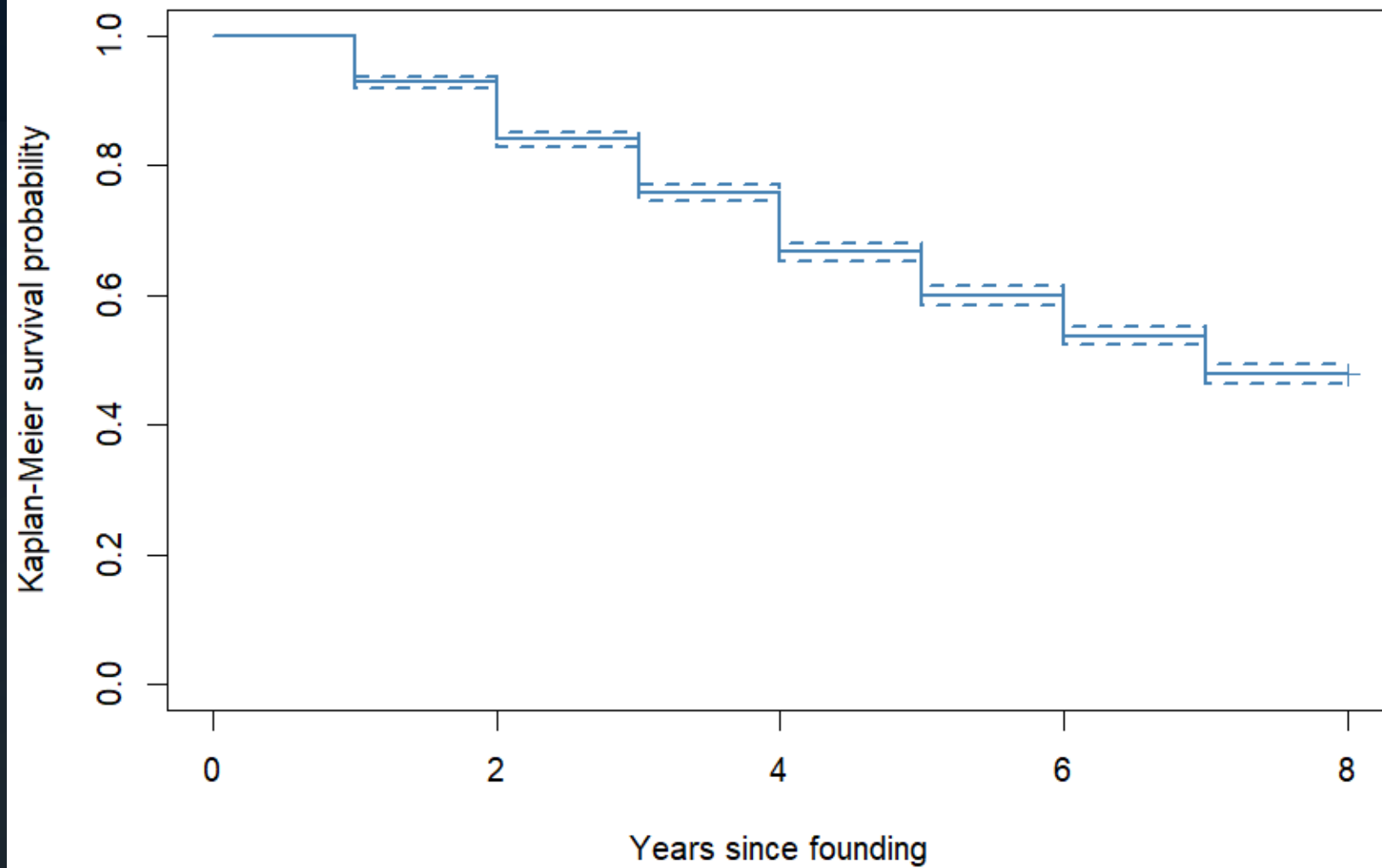
```
graph TD; A[Excluded firms that dropped out of study] --> B[System-missing codes recoded to 0 across predictor columns]; B --> C[Added missing indicator flags for key variable];
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The diagram illustrates a three-step data cleaning process. It begins with a light blue box at the top, followed by a medium blue box in the middle, and ends with a dark blue box at the bottom. Each step is connected to the next by a downward-pointing arrow, indicating a sequential workflow.

System-missing codes recoded to 0
across predictor columns

Added missing indicator flags for key
variable

Kaplan-Meier survival curve: all firms



Why LASSO?

- Uses feature selection to identify predictors that are strongly associated with survival rate
 - Large number of predictors relative to the sample size
- Reduces overfitting and potential for masking important predictors
- Addresses multicollinearity and highly correlated variables
- Carried into final Cox model

LASSO Details

Details

10-fold cross-validation
1-SE rule to select final penalty
Bootstrapping (50 resamples)

Results:

208 predictors → 19 candidate predictors

Why Bootstrapping?

Problem that Bootstrapping solves:

- Correlated variables can cause LASSO to choose different predictors depending on the sample

Process

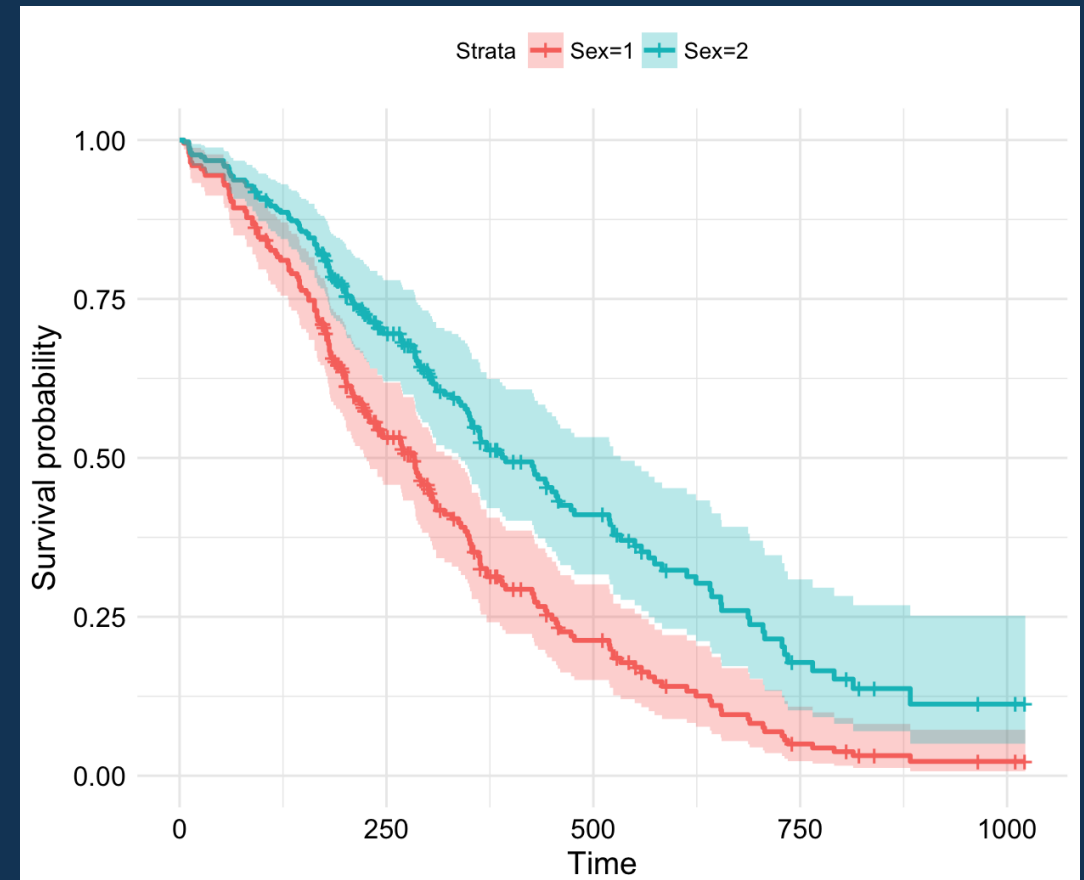
1. Sample businesses with replacement
2. Fit LASSO
3. Record selected variables
4. Repeat 50 times

Results

- Only predictors selected consistently were kept

Cox Proportional Hazards Model

- A **survival analysis** model that **predicts** the risk of an event occurring over time based on predictor variables
- **Produces:**
 - Hazard ratios (HRs) for each predictor
 - An overall risk score for each business
 - A survival curve per business showing the probability the business survives over time
- **Unlike traditional prediction models, the Cox model predicts:**
 - Whether a business is likely to exit
 - When that exit is most likely to occur



Our Model

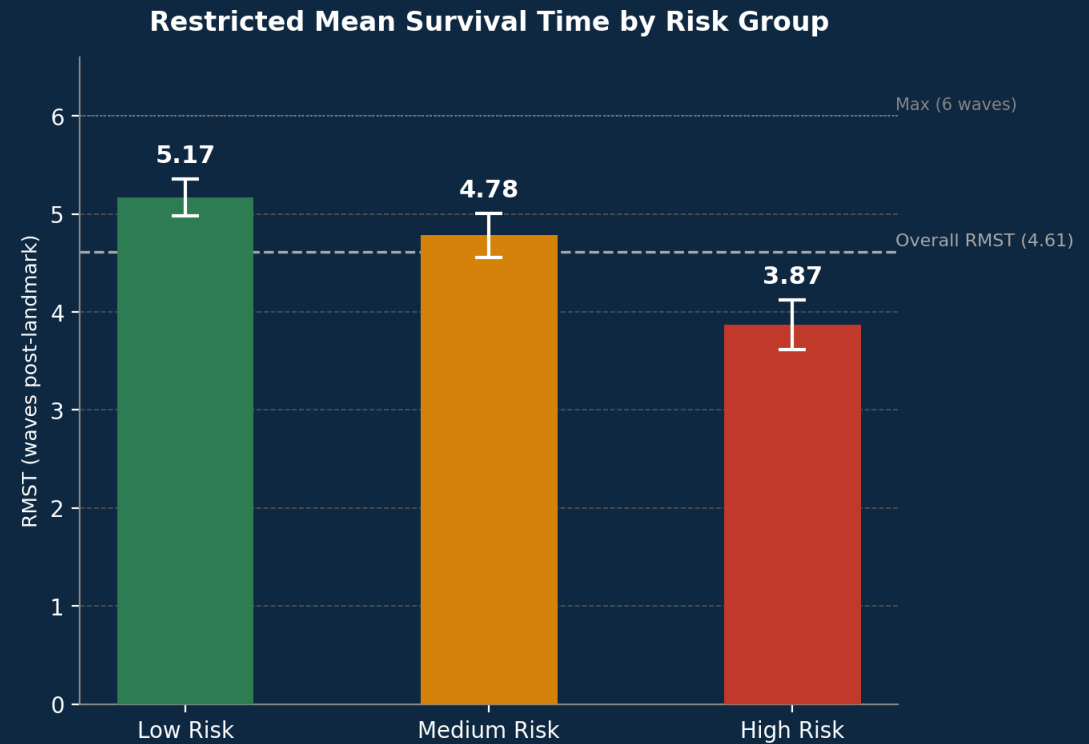
- **Landmark Cox Proportional Hazards Model**
- **What it does:**
 - Predicts if and when businesses will exit
 - Estimates each business's probability of surviving over time
 - Produces individual survival curves for each business
 - Outperformed other models:
 - Random Survival Forest
 - DeepSurv
 - Discrete-time Neural Net

Why Landmark?

- **Landmark:**
 - Landmark is fixed point in time where we collect each business's information and begin predicting what happens after that point
- **Benefits:**
 - Prevents leakage
 - All businesses evaluated from same point
- **Why Wave 2?**
 - Businesses have established operating history
 - We excluded all businesses exiting before wave 2 from our final training dataset

Restricted Mean Survival Time

- Restricted Mean Survival Time
- Average time a firm is expected to survive up to a fixed horizon



Model Performance Indicators

Concordance Index (C-index):

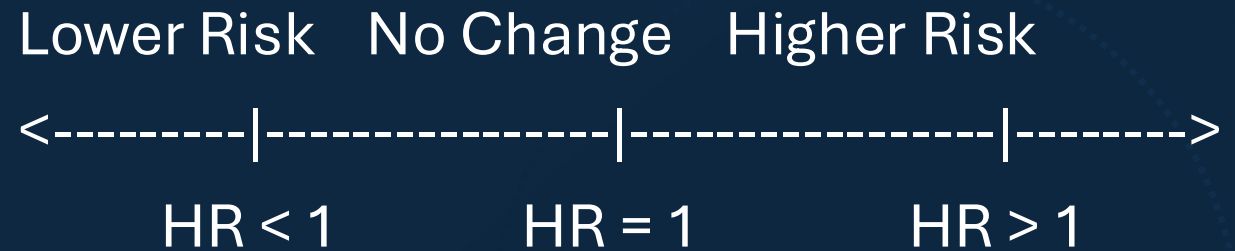
- For every pair of businesses, how often does the model correctly identify which one exits sooner?
- 0.5 = coin flip | 1.0 = ideal

Results:

- Training C-index = 0.672
- Testing C-index = 0.661
- Gap of 0.011 between testing and training c-indexes — minimal evidence of overfitting

Model Outputs

- **Hazards Ratio (HR):**
 - Measures which business characteristics were most closely associated with survival



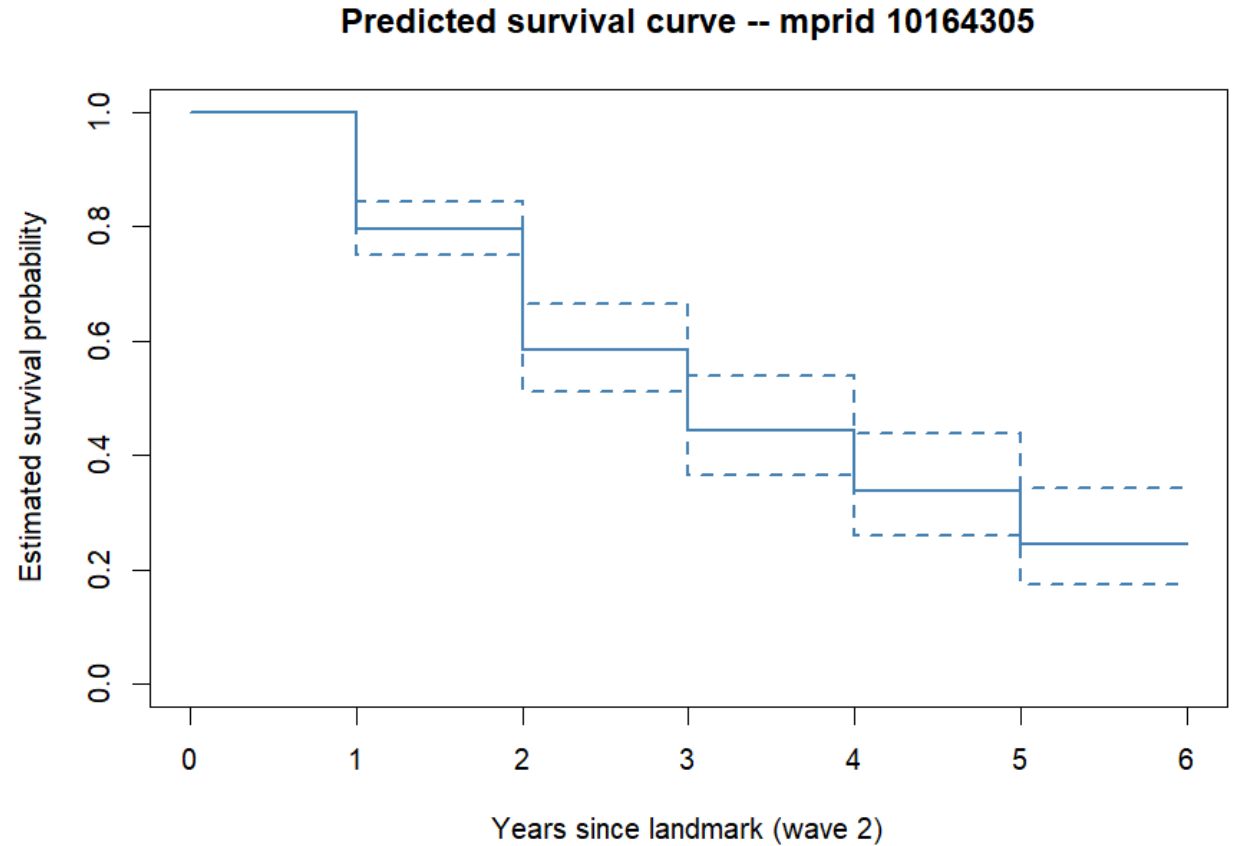
- **Survival Curve:**
 - A predicted Survival Curve for each individual business from the testing dataset

Model Output - HR

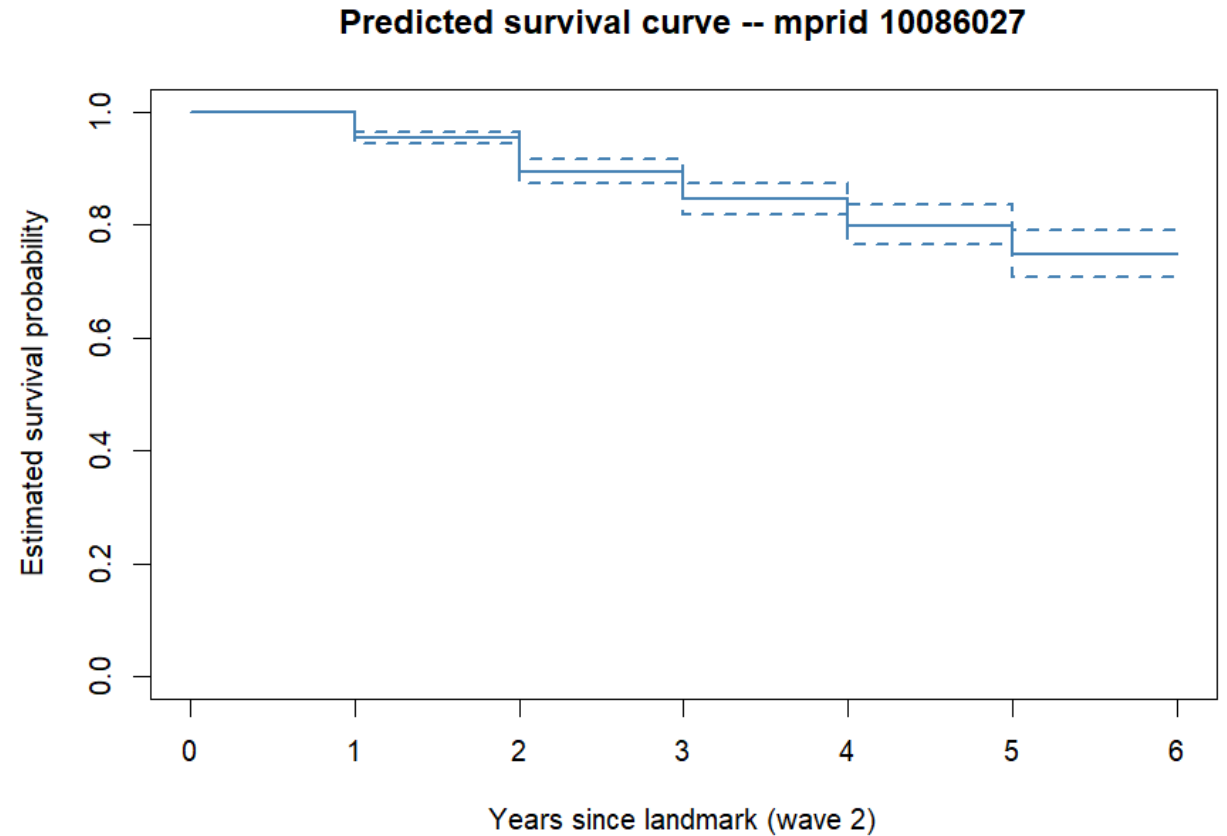
Most Significant Predictors →

Predictor	Hazard Ratio	C-Interval
Cash Assets	0.919	0.893 – 0.945
Started by Acquiring a Business	1.78	1.266 – 2.501
Corporation Legal Status	1.296	1.049 – 1.599
Equity Data Missing	3.79	1.323 – 10.841
Credit Card Data Missing	1.18	1.003 – 1.387
Credit Risk Missing	1.229	1.018 – 1.482

Model Output - Predicted Survival Curve



Model Output - Predicted Survival Curve



Limitations

- 46.8% of cells are missing
 - Missing data
- Single 2004 cohort
- C-index of 0.66—real signal, but substantial unexplained variation remains

AI usage

- Debugging
- Assistance with building code
- Better conceptual understanding of our model



Max Gebauer



To:  Chen, Zixuan

Fri 7/10/2026 10:36



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Extremely proud of you all. Excellent presentation and conceptual understanding demonstrating